3 A line joins A (1, 3) to B (5, 8).

(a) **(i)** Find the midpoint of AB.

(..... , ...) [2]

(ii) Find the equation of the line AB.
Give your answer in the form $y = mx + c$.

$y =$ ..... [3]

(b) The line AB is transformed to the line PQ.

Find the co-ordinates of P and the co-ordinates of Q after AB is transformed by

(i) a translation by the vector $\begin{pmatrix} 5 \\ -2 \end{pmatrix}$,

P (..... , ...)

Q (..... , ...) [2]

(ii) a rotation through 90° anticlockwise about the origin,

P (..... , ...)

Q (..... , ...) [2]

(iii) a reflection in the line $x = 2$,

P (..... , ...)

Q (..... , ...) [2]

(iv) a transformation by the matrix $\begin{pmatrix} -1 & 2 \\ 0 & -1 \end{pmatrix}$.

P (..... , ...)

Q (..... , ...) [2]

(c) Describe fully the **single** transformation that maps the line AB onto the line PQ where P is the point $(-2, -6)$ and Q is the point $(-10, -16)$.

.....

..... [3]